INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

SPRAYERS, INSECTICIDE, MANUALLY-CARRIED: HAND-OPERATED-COMPRESSION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description covers manually carried, hand- operated-compression insecticide sprayers. The sprayers are intended for manual spraying of insecticides and fungicides.

2. CLASSIFICATION. The sprayers shall conform to the following sizes. Operating capacity is defined as maximum liquid fill of the tank when first used - 3/4 liquid fill and 1/4 air capacity with all accessories in place:

2.1 Size.

Size 1 - 1 gallon operating capacity.

Size 2 - 2 gallon operating capacity.

3. SALIENT CHARACTERISTICS.

3.1 Description. The sprayers shall be of the cylindrical type and equipped with a hand-operated air pump for pressurization, hose, cut-off valve, lance, nozzle and other accessories as specified herein.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to DLA Aviation, VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil.

AMSC N/A FSC 3740

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3.2 Materials. All materials shall be resistant to common insecticides and fungicides formulations, oils and fungus.

3.2.1 Tank. The tank, including the removable filler hole cover and any fittings, shall be entirely of metallic construction. The metal may be corrosion resistant steel (CRES), copper, brass or bronze. Joining materials or welds and heat-affected zones shall be corrosion resistant.

3.2.2 Other components. All other components shall be of design and material capable of withstanding normal usage conditions.

3.3 Dimensions. The maximum height of the tank from the bottom of the base to the uppermost point of the top shall not exceed 11 inches (27.9 cm) for size 1 and 22 inches (55.9 cm) for size 2. The height from the base of the sprayer to the handle in its lowest position shall not exceed 6 additional inches (15.2 cm) above the height of the tank. The diameter of the tank shall not exceed 9 inches (22.9 cm).

3.4 Weight. The weight of the sprayer, when empty but equipped for operation, shall not exceed 8 pounds (3.6 kg) for size 1 and 10 pounds (4.5 kg) for size 2.

3.5 Tank markings. The tank shall be marked in a legible and permanent manner with horizontally embossed lines to indicate the fill charges. The fill marks shall appear at 0.5 gallon (2L) intervals. The maximum fill charge shall be at 3/4 of the total tank capacity. Other markings shall indicate the year of manufacture.

3.6 Tank fittings.

3.6.1 Fastenings for lance. The tank shall be fitted with a pair of fastenings to secure the nozzle and lance and when specified (see 7.2.c), extra fastenings for an extension lance. Fastenings shall withstand dropping to the extent that they may be bent back to their original shape and remain functional. No set screws shall be used for fastenings.

3.6.2 Filler hole. The filler hole shall be not less than 3.7 inches nominal (9.4 cm) in diameter if circular, or in minor axis if oval. It shall have no sharp edges. The filler hole cover shall be attached to the tank with a chain capable of withstanding a breaking strain of not less than 110 pounds (50 kg).

3.6.3 Riveted fittings. Rivets used for securing lance brackets or other external fittings shall not penetrate the pressurized portion of the tank.

3.6.4 Pressure-relief (safety) valve. The tank shall be equipped with a fixed pressure-relief valve for whenever an air-inlet valve is used to prevent pressurization in the tank beyond the limits prescribed for safety or operational purposes. Threaded fittings subject to pressure shall be channeled. Relief pressure is at manufacturer's option but shall not exceed 100 psi (700 kPa). Relief pressure shall be permanently marked on the valve.

3.6.5 Pressure gauge opening. A fitting for installation of a pressure gauge shall be provided on the tank body. When a pressure gauge is not required (see 3.16.11 and 7.2.7) a plug shall be provided

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for the opening into which the gauge would fit. The plug shall provide an air-tight and liquid-tight closure.

3.7 Pump.

3.7.1 Construction. The pump cylinder shall be a separate part, independently detachable from the tank, tank top, or any other component of the sprayer. The pump shall be easily disassembled and reassembled for cleaning or maintenance.

3.7.2 Check valve. The pump shall include a check valve that is air-tight and liquid-tight. It shall be so designed as to prevent any leaking of liquid into the pump cylinder over a period of 15 minutes when the tank is pressurized between 65-70 psi (448 - 483 kPa) and the check valve is fully submerged.

3.7.3 Pump operation. The pump shall pressurize the tank on the down stroke of the plunger. The pump shall provide a minimum pressure of 80 psi (552 kPa) or the maximum tank operating pressure, whichever is lower, in the tank.

3.7.4 Plunger shaft. The plunger shaft shall be of steel and may be either a solid rod or a tube.

3.7.5 Handle. The pump handle shall have a grip of not less than 3.56 inches (9.0 cm). The pump handle shall be so designed that a complete stroke of the pump may be obtained with the pump handle turned to any position. The compression strength of the assembly, including the handle, its components, and its junction with the plunger rod, shall not be less than that of the plunger rod. Whatever the material of manufacture, it shall be nonabsorbent.

3.7.6 Handle locking device. The plunger shaft shall be fitted with a quick-acting locking device that, when applied, will secure the plunger assembly at the lowest operating position for carrying.

3.8 Dip tube. The dip tube shall extend to within approximately 0.4 inch (10.2 mm) of the bottom of the tank. The end of the tube within the tank shall be cut at an angle to prevent blockage. The dip tube shall be securely supported at the point where it emerges from the tank.

3.9 Strainers. Two strainers shall be provided. The two strainers shall be of the same type. One strainer shall be in the discharge line between the entry to the dip tube and the entry ports of the cut-off valve. One strainer shall be in the cut-off valve. The strainers shall have sufficient structural strength or be mounted in such a way that they will not be subject to accidental puncture or collapse. They shall be easily removable and replaceable without special tools. The strainers shall allow easy passage of liquid.

3.10 Hose. Unless otherwise specified (see 7.2.d), the hose shall be of a material resistant to common insecticidesand fungicides formulations or oils. The hose shall have an operating pressure of at least 100 psi (690 kPa). A length 5 feet (1.5 m) with an inside diameter between 0.250 to 0.375 inch (6.4 to 9.5 mm) shall be provided with each sprayer. The hose shall be uniform in size and free from defects. .

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3.11 Hose connections. Hoses shall be retained on connections or couplings using removable/reusable clamps or threaded connections. The clamps shall have no sharp edges and a minimum of extended projections liable to catch on clothing or other objects. Threaded connections shall have finished, hexagonal faces or opposing flattened surfaces and provide air-tight and liquid-tight joints at the highest designated operating pressure when installed using only hand force. If any tools are required for disassembly/assembly, only a wrench and/or flat screwdriver shall be needed.

3.12 Cut-off valve.

3.12.1 Construction. The valve shall be so constructed that inner and outer parts are readily accessible for frequent cleaning without the use of special tools. The valve shall conveniently fit the hand and shall be free of sharp edges or projections

3.12.2 Connections. Valves shall be equipped with screw threads at inlet and outlet ports. The outer surfaces of the valve, adjacent to the screw threads, shall have opposing faces, to permit the use of a wrench, if tools are required for its removal. Knurled faces or wing nuts of adequate shape and size may be used provided that the valve can be loosened and tightened adequately by the pressure of thumb and fingers alone, without the use of tools, at the highest designated operating pressure.

3.12.3 Valve operation. Valve operation may be by lever, push button or other easily operated means. The force required to operate the valve shall not exceed 10 lbf (44 N).

3.12.4 Leakage. Cut-off valves shall not drip or leak through the valve or around the packing when subjected to normal operating pressures over a period of one year.

3.13 Lance. The lance shall be easily detachable/re-attachable and provide an air-tight and liquid-tight joint at the highest operational pressure using only hand-force or a wrench. It shall have the same thread and size at both ends so that any number of lances in series, or the nozzle alone, may be connected directly to the cut-off valve housing. Standard lances shall be between 16 and 20 inches (40-50 cm) in length, measured between the external faces of the joints.

3.14 Nozzle.

3.14.1 Description. The spray nozzle shall be of the hydraulic-energy type in which droplets are formed by forcing the spray material through an orifice. It shall consist of a droplet-producing nozzle tip plus a nozzle body and nozzle cap. The nozzle tip shall be removable and replaceable without the use of tools, and shall not leak after reassembly.

3.14.2 Nozzle assembly. The nozzle assembly consisting of the body, cap and tip shall be constructed from stainless steel, ceramic or brass.

3.15 Gaskets. Where possible, gaskets shall be wholly or partially recessed into the parts into which the joints are made.

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3.16 Optional items. When required, the following items shall be specified (see 7.2.g):

3.16.1 Air-inlet valve. An air-inlet valve (automobile type) for the purpose of supplying air pressure to the sprayer from an outside pump or high-pressure source. With such an inlet valve, a highly reliable pressure/relief (safety) valve shall be required (see 3.6.4).

3.16.2 Regulated delivery system. A regulated flow-delivery system, consisting of a complete nozzle assembly and disc flow regulator. The discharge rate and distribution pattern shall be as specified (see 7.2.h).

3.16.3 Constant-pressure regulator. The sprayer may be equipped with a constant-pressure regulator inserted in the discharge line, adjustable to produce tip pressures in the program operating range, as specified (see 7.2.i).

3.16.4 Lances. Extra lances for use as extensions. These may be the same length as furnished with the sprayer or as specified (see 7.2.j). Leak-proof telescopic lances may be provided (see 7.2.k).

3.16.5 Swivel nozzles. Leak-proof swivel nozzles that can readily be rotated by hand. Additional swivel nozzles shall be as specified (see 7.2.l).

3.16.6 Multiple spray nozzle. A multiple spray nozzle which has an adjustable cone spray and straight stream application.

3.16.7 Gooseneck attachment. A gooseneck attachment consisting of a section of lance, curved 120 degrees, with 8 inches (20cm) length of tubing, with a coupling on each end to permit attachment to the lance nozzle body.

3.16.8 Gooseneck lance. A nine inch (23 cm) gooseneck lance with an extra 20 inch (50.8 cm) extension lance. Additional fastenings will be required on the tank for the attachment of the lance.

3.16.9 Nylon nozzle body. A nozzle body and cap made of shockproof nylon (containing not less than 2 percent moisture by weight) in place of the normal brass fitting.

3.16.10 Pressure gauge. A back-mounting pressure gauge with a scale reading from 0 to 100 psi, O to 700 kPa or both. The gauge shall not be affected by the deposition and accumulation of spray materials. The dial cover shall be shatterproof and moisture-proof and permit easy reading of the tank pressure.

3.16.11 Foot-rest. A foot-rest designed to steady the apparatus while it is pumped. It shall be attached to the bottom part of the tank and shall be hinged to fold up against the tank side and remain there until required.

3.16.12 Inspection light. A spray inspection light. It shall be attached to the lance

3.16.13 Crack and crevice tip assembly. A crack and crevice nozzle tip assembly.

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3.16.14 Treatment and finish. Unless otherwise specified (see 7.2.m), the sprayer shall be treated and finished in accordance with the quality practiced by the manufacturer to obtain treatment and finish of equivalent commercial units.

3.17 Identification marking.

3.17.1 Civil identification marking. The sprayer shall be marked in a permanent and legible manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.17.2 Military identification marking. The sprayer shall be identified in accordance with MIL-STD-130.

3.18 Workmanship. All parts, components, and assemblies of the sprayer shall be clean and free from harmful extraneous material such as sand, dirt, fins, pits, scale, and rust, and shall also be free from burrs and sharp edges or projections or other defects that could impair its performance or service- ability.

3.19 Operator's manual. The manufacturer's standard instruction manual with parts lists shall be provided with each sprayer.

4. REGULATORY REQUIREMENTS.

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. PRODUCT CONFORMANCE PROVISIONS.

5.1 Product conformance. The product provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Purchase approval. The purchaser should obtain Major Command pest management professional consultant approval before purchase.

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7.2 Ordering data. The contract or order should specify the following:

a. Number, title, and date of this commercial item description.
b. Size of sprayer required.
c. When extra fastenings will be required for extension lance (see 3.6.1).
d. When hose material other than as specified is required (see 3.10).
e. Optional items to be included with sprayer (see 3.16).
f. The discharge rate and distribution pattern when a regulated delivery system is required (see 3.16.2).
g. When a constant-pressure regulator is required (see 3.16.3).
h. The number and length of extra lances required (see 3.16.4).
i. When a leakproof telescopic lance is required (see 3.16.4).
j. When additional swivel nozzles are required (see 3.16.5).
k. When treatment and finish will be other than the manufacturer's equivalent commercial units (see 3.16.15).

7.3 Source of Documents.

7.3.1 Electronic copies of MIL-STD-130 are available online at http://quicksearch.dla.mil. .

7.3.2 Electronic copies of the FAR are available online at https://www.acquisition.gov.

7.4 National Stock Numbers (NSNs). The following is a list of NSNs assigned which correspond to this CID.

NSN	Size
3740-00-191-3677	1
3740-00-641-4719	2

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MILITARY INTERESTS:

CIVIL AGENCY COORDINATING ACTIVITY:
Custodians:
Army – MD
Navy – YD
Air Force – 99
DLA – GS

Review Activities:
Navy - CG, MC, MS
Air Force – 84

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

PREPARING ACTIVITY:

DLA – GS3

Project Number 3740-2017-001

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